Paper replication

Post-Earnings-Announcement Drift: Delayed Price Response or Risk Premium?

Victor L. Bernard and Jacob K. Thomas

Journal of Accounting Research, 1989, Vol. 27
Current Studies on Information Content of Accounting Earnings (1989), pp. 1-36

1 Research question, methodology and key findings

Bernard and Thomas (1989) investigate the post-earnings-announcement drift (PEAD) anomaly, characterized by the continued directional movement of stock prices following earnings announcement. They seek to distinguish whether this drift reflects delayed market response to earnings information or compensation for underlying risk factors not adequately captured by prevailing asset-pricing models.

The paper uses an event-study approach on 84,792 NYSE/AMEX firm-quarters (1974-1986). The principal result reveals clear evidence of PEAD: a long-short strategy in extreme standardized unexpected earnings (SUE) deciles earns 4.2% over the 60 post announcement days. Drift is stronger among smaller firms, suggesting size-related factors play a role (investor attention, trading costs). Most drift occurs within 60 days, with little evidence of significant drift beyond 180 days. The authors then test the risk-premium explanation. *Beta* shifts (Ball et al., 1988) are confirmed, but far below values required to explain the drift magnitude. APT risk factors (Chen et al., 1986) show no significant correlation with SUE strategy returns. Raw returns on bad news firms are difficult to reconcile with risk: their near zero or negative returns in the first five post announcement days are inconsistent with these being risky assets requiring compensation. Profitability of the strategy in over 80% of quarters further undermines the risk explanation. Transaction costs are also examined: PEAD returns appear bounded, yet mixed results provide weak support for short-sale constraint explaining the anomaly. Finally, the authors show that current earnings news predicts the market reaction to the next quarter's earnings announcement. Based on SUE, good (bad) news firms again have positive (negative) abnormal returns. This suggests investors underreact to earnings persistence, then respond when next quarter's earnings confirm the signal.

Overall, Bernard and Thomas conclude that PEAD is difficult to explain as a risk premium. The evidence is more consistent with delayed price response, particularly the failure of investors to fully understand what current earnings imply for future earnings.

2 Empirical results replications

2.1 Data

The authors study NYSE and AMEX firms over 1974-1986, drawing on CRSP and Compustat data. We use Wharton Research Data Services (WRDS) as our data source (Table 1). Following the paper's methodology, we select firms having at least ten consecutive observations in the 1982-1987 Compustat files, that are present in the CRSP Daily file and listed on NYSE/AMEX[1]. Observations are defined as quarters having valid diluted earnings per share excluding extraordinary items[2].

TABLE 1. Dataset Summary

Dataset	Query parameters	Period	Rows	Purpose
CRSP Daily	primaryexch, permno, ticker, dlycap, dlyret	1974, 1986	18.36M	Sample
CRSP Daily	primaryexch, permno, ticker, dlycap, dlyret	2012, 2024	25.62M	OOS
Compustat Daily	tic, rdq, fyearq, fqtr, prccq, epsfxq, epspxq	1961, 1987	0.37M	Sample
Compustat Daily	tic, rdq, fyearq, fqtr, prccq, epsfxq, epspxq	1999, 2025	1.06M	OOS
CRSP/Compustat	gvkey, tic, cusip, permno, linkdt, linkenddt	None	0.04M	Link

2.2 Metrics

2.2.1 Abnormal returns

Abnormal returns are computed as:

$$AR_{jt} = R_{jt} - R_{pt}$$

where R_{jt} is the raw return for firm j, day t, R_{pt} is the equally weighted mean return for day t on the NYSE firm size decile of firm j and AR is abnormal returns of firm j on day t.

Size deciles are defined from market capitalization on the first trading day of the year of all NYSE/AMEX firms[3]. Observations are excluded if returns on earnings announcement day or in the 160 days surrounding it are missing[4].

[1]The authors also select firms from NASDAQ. However, due to the limited use of this data in later parts of the paper, we decided not to include this data in our own replication.

[2]For earnings per share excluding extraordinary items, the paper doesn't mention if diluted (epsfxq) or basic (epspxq) feature should be used. After comparison over the full pipeline, we selected epsfxq: despite being equivalent in almost every aspects, it brought us closer to the shortlisted quarters count of the initial paper.

[3]For newly listed firms, we use the earliest daily market cap in that year

[4]No additional information about this window is mentioned: we assume a symmetrical ±80 days window.

2.2.2 Estimation of standardized unexpected earnings (*SUE*)

Earnings are forecasted using Foster's 1977 model, assuming that earnings follow a first-order autoregressive process based on previous year earnings and adjusted seasonal differences.

$$E(Q_t) = Q_{t-4} + \phi_1(Q_{t-1} - Q_{t-5}) + \delta$$

Where Q_t are quarter t earnings, ϕ_1 is the autoregressive parameter and δ is a drift term.

We exclude firms with fewer than ten consecutive quarters, assume a seasonal random walk for those with less than 16 observations and use a maximum of 24 observations for estimation. *SUE* is the residual scaled by the standard deviation of errors over the estimation period. After fulfilling *SUE* requirements, we obtain 89,226 firm-quarters: 4,434 more than the original. This gap likely reflects modern WRDS Compustat's retroactive corrections and backfilling, which differ from the vintage tapes used by the authors.

2.2.3 Continuously balanced *SUE*

The *SUE* approach is difficult to implement in practice, requiring daily rebalancing of hundreds of stock. The continuously balanced *SUE* tests the sensitivity of results to this is*SUE*.

Each day, we select firms announcing earnings whose *SUE* falls in the extreme quintiles of the prior-quarter distribution. When both exist, we take an equally weighted long-short position canceling each side's amount and measure returns over 60 trading days. This matching process is conducted within size groups. If no match can be created on a trading day, we "wait" until a match is created.

2.3 Empirical results[5]

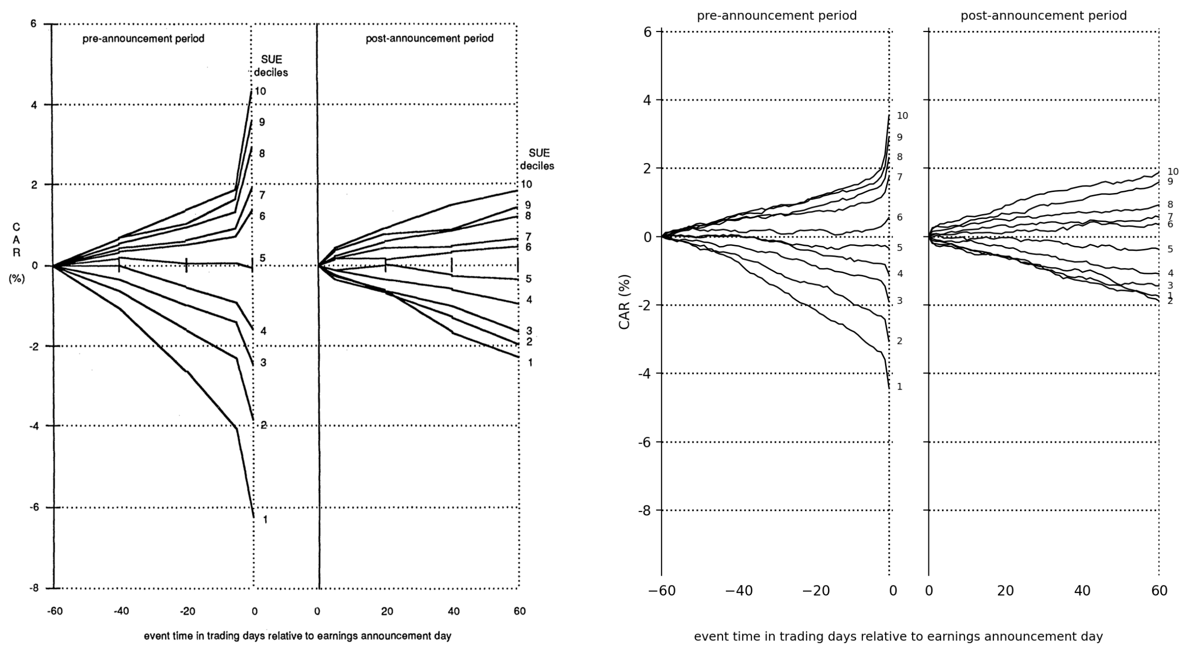
2.3.1 Descriptive results

a) Magnitude of the drift

Cumulated abnormal returns (CARs) are computed over pre/post announcement periods and assigned to portfolios based on prior-quarter *SUE* rankings. Our results closely replicate the original: *SUE* decile ordering is almost perfectly preserved and we notice a similar sharp CAR change the day before earnings announcement (Figure 1). Our replication matches the authors' returns within ±15% (Table 2).

[5] We do not replicate APT risk factor regression, raw returns on bad news firms, nor tests of transactions costs as explanation for drift.

Figure 1. Cumulative returns (CARs) for *SUE* portfolios: all announcements.



(a) Original paper (b) Replication

b) Relation of drift to firm size

We replicate the original tighter (wider) PEAD for larger (smaller) firms. We omit in-sample plots as they resemble out-of-sample results. Returns closely match the original for small firms but diverge for large firms (Table 2). The main finding holds: drift decreases as firm size increases.

Table 2. *SUE* Long/Short Returns Comparison

	***SUE* 60d (%)**				**Cont. *SUE* 60d (%)**			
Source	**All**	**Small**	**Medium**	**Large**	**All**	**Small**	**Medium**	**Large**
Paper	4.2	5.3	4.5	2.8	3.84	5.1	4.3	2.8
Replication	3.63	5.14	3.73	1.94	3.99	5.82	4.19	2.1
	(-0.57)	(-0.16)	(-0.77)	(-0.86)	(+0.15)	(+0.72)	(-0.11)	(-0.70)

c) Longevity of the drift

Assuming that drift is fully realized within 480 trading days, most of it occurs within the first 60, with little statistically significant drift beyond 180 days (Table 3). The 60-day drift as a fraction of the full drift is 50%, 67%, and 54% (S/M/L): comparable to the paper's 53%, 58%, and 76%.

Table 3. Longevity of Post-Earnings-Announcement Drift

Cumulative abnormal returns (CAR) for high and low SUE portfolios

Holding Period (Trading Days)	Small Firms Diff (Hi-Lo)	Medium Firms Diff (Hi-Lo)	Large Firms Diff (Hi-Lo)
Postannouncement Drift (Cumulative Abnormal Returns, %)			
1 to 60	5.19	3.71	1.92
1 to 120	7.46	5.49	3.34
1 to 180	10.01	6.64	3.21
1 to 480	10.46	5.53	3.55
Postannouncement Drift (as a Fraction of 480-Day Drift)			
1 to 60	0.50	0.67	0.54
1 to 120	0.71	0.99	0.94
1 to 180	0.96	1.20	0.90
1 to 480	1.00	1.00	1.00

2.3.2 Test of risk premiums as explanation for the drift

a) Shifts in betas as a potential explanation

Following Ball et al. (1988), we address *beta* shifts by pooling compounded excess returns for firm-quarters within each *SUE* decile and window. We regress against compounded excess market returns using the CAPM formula and additional data from French's website:

$$R_{jt} - R_{ft} = a + b(R_{mt} - R_{ft}) + e_{jt}$$

The positive *SUE-beta* relation across windows matches the paper: *beta* shifts exist around earnings announcements, yet they fall far short of explaining PEAD's magnitude (Table 4).

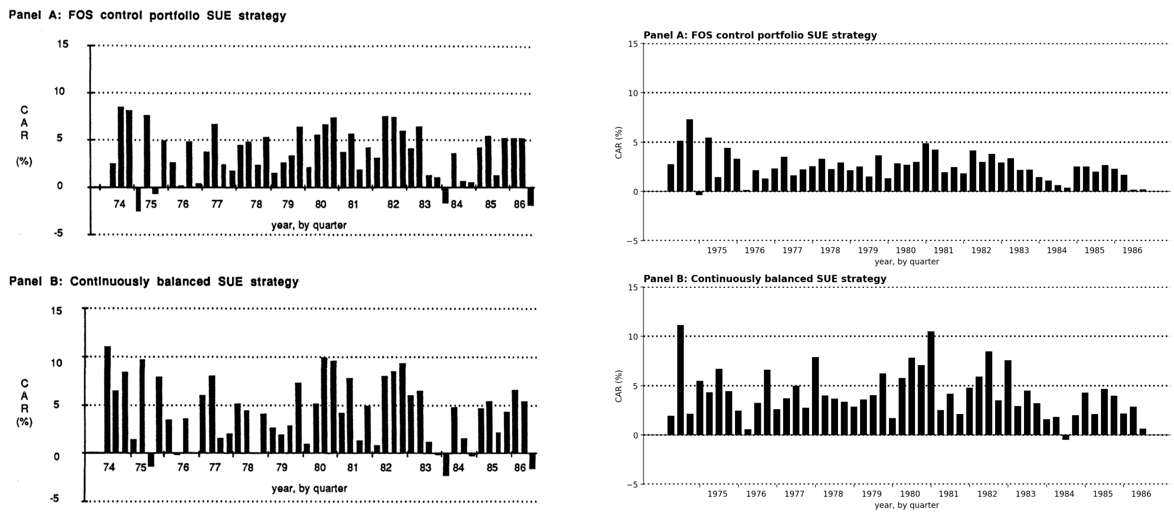
Table 4. *Beta* Estimates by *SUE* Category, in Periods Surrounding Earnings Announcement

	Preannouncement Period		Postannouncement Period			
SUE Decile	(-119, -60)	(-59, 0)	(1, 60)	(61, 120)	(121, 180)	(181, 240)
Beta estimates						
1	1.16	1.21	1.14	1.17	1.26	1.31
2	1.20	1.19	1.18	1.15	1.19	1.25
9	1.24	1.29	1.27	1.29	1.20	1.18
10	1.25	1.35	1.29	1.26	1.21	1.15
Rank correlation, *SUE* & *beta*	0.70*	0.89*	0.90*	0.72*	-0.22	-0.85*
Jensen's alpha						
SUE = 1	-1.6	-4.0*	-0.8	-0.6	-0.1	0.8
SUE = 10	2.8	6.2*	3.2	1.5	1.2	1.3
Combined	4.4	10.2*	4.0	2.0	1.4	0.4

b) Consistent profitability of the strategy

We define two strategies: A (long/short on top/bottom SUE quintiles) and B (cont. SUE), with 60-day CARs allocated to calendar quarters by the fraction generated within each one. Both generate consistently positive returns (Figure 2). Authors report 46/50 positive quarters (A) and 44/50 (B). We observe only one negative quarter per strategy, with few near-null quarters. Magnitudes are comparable, confirming PEAD as a strong and persistent anomaly.

Figure 2. Cumulative abnormal returns (CARs) from SUE strategies, by calendar quarter.



(a) Original paper (b) Replication

2.3.3 Test of whether future earnings reflect full implication of past earnings

To test whether past earnings (quarter t) predict market reactions to quarter t+1 earnings announcement, we use extreme SUE deciles from t to measure abnormal returns over days (-4, 0) surrounding t+1 announcements. Under market efficiency, past information should not be able to predict abnormal returns.

Our D1/D10 differences as a fraction of 60-day drift is close to the original 40%, 29% and 25% - though the (-4, 0) window represents only 8% of the period.

Table 5. Mean Stock Price Reaction to Quarter t+1 Earnings (Firms Grouped on Quarter t SUE)

	Abnormal Returns in [-4,0] Window of Q t+1 Earnings (in %) (t-values)		
SUE Decile for Quarter t	Small	Medium	Large
10 (good)	1.54 (9.61)	0.82 (6.84)	0.47 5.85
1 (bad)	-0.49 (-3.28)	0.42 (-3.54)	-0.35 (-4.33)
Difference	2.03	1.24	0.82
As fraction of 60-day drift	39%	33%	43%

3 Out-of-sample extension

3.1 Data

For out-of-sample replication, we follow all procedures aforementioned to obtain 83,836 firm-quarters using CRSP files from 2012 to 2025 and Compustat files from 1999 to 2025.

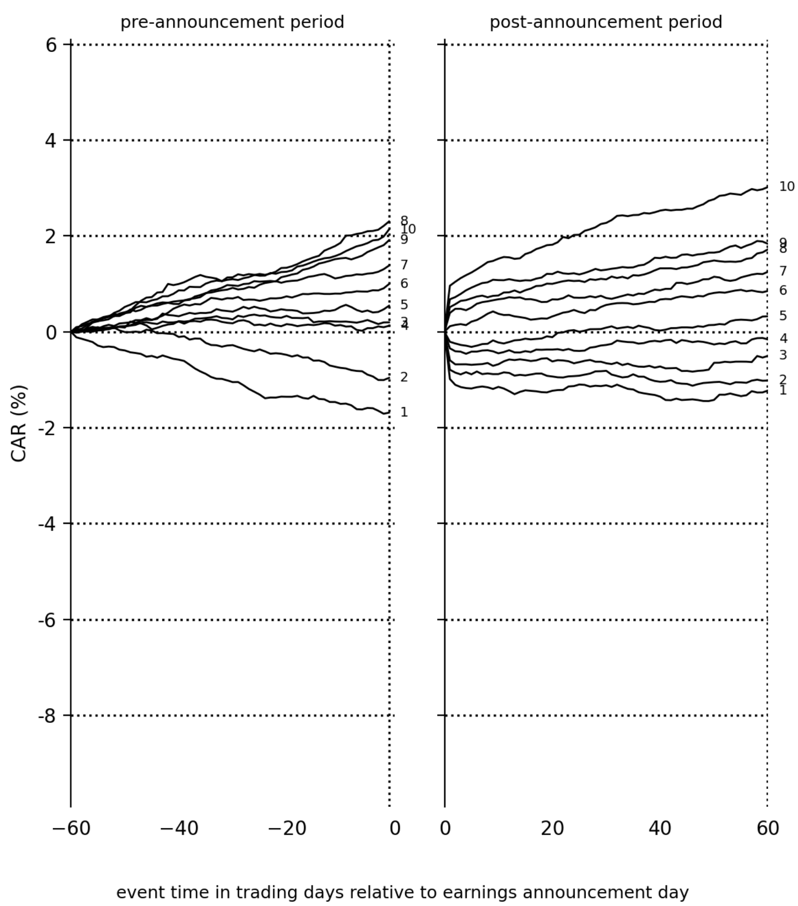
3.2 Empirical results

3.2.1 Descriptive results

a) Magnitude of the drift

PEAD remains present OOS (Figure 3). However, the sharp price movement is this time located after the earnings announcement. A long-short strategy in extreme *SUE* deciles yields 4.24% over 60 days (19.07% annualized, 14.89% for the cont. *SUE*), comparable to the original sample.

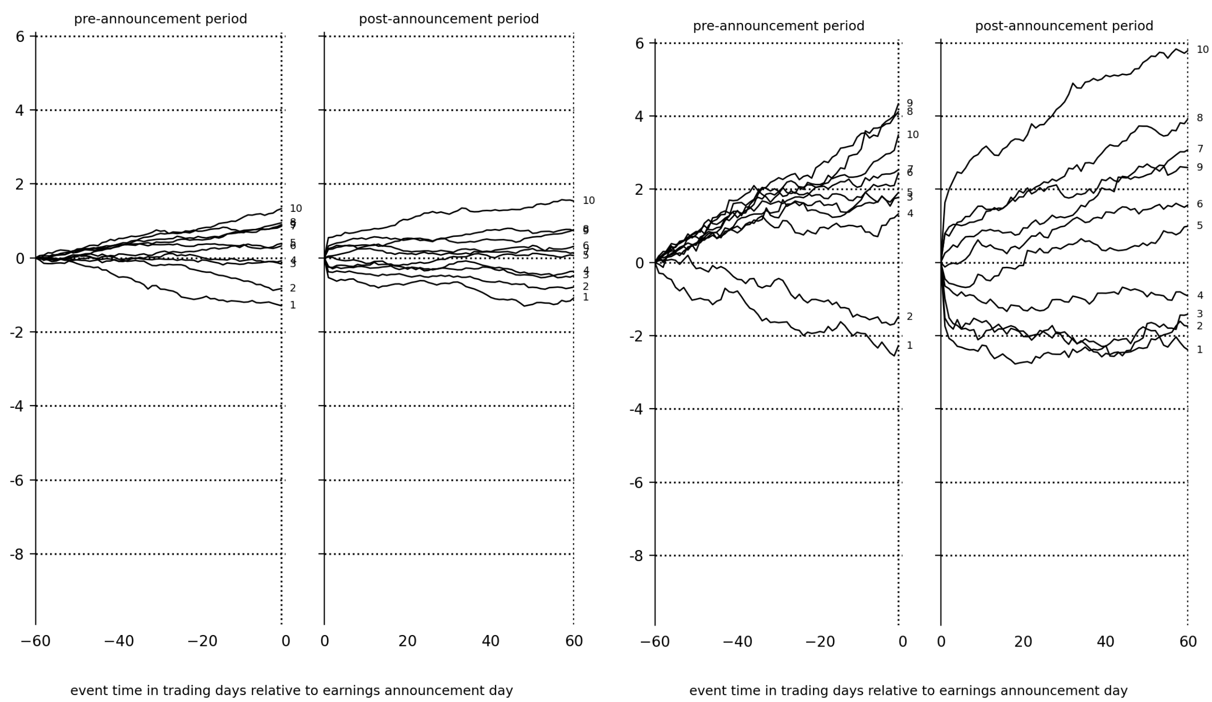
FIGURE 3. Out-of-sample cumulative returns (CARs) for *SUE* portfolios: all announcements.



b) Relation of drift to firm size

The size effect persists, with tighter spread for large firms and wider drift for smaller firms (Figure 4). Returns still decrease as firm size increases, for both strategies. 60-day *SUE* yields 39.20% and 11.41% annualized returns (S/L) (cont. *SUE* yields 32.49% and 7.17% annualized respectively).

FIGURE 4. Out-of-sample cumulative returns (CARs) for *SUE* portfolios: by firm size.



(a) Large firms only (b) Small firms only

c) Longevity of the drift

Longevity reveals a structural change. Sample's 60-day drift was 50% to 67% of the 480-day drift. OOS, these figures are 95%, 142%, and 301% (S/M/L). Drift is fully realized within 60 days (Table 6). Analyzing shorter horizons (one, five and ten days) reveals that day one drift represents 40%, 87%, and 128% (S/M/L) of the full PEAD, reaching 55%, 93%, and 150% in five days. For medium firms, long-term price adjustment takes ten trading days, followed by long-term reversal. Unlike the original sample where drift took several months, markets now appear to incorporate earnings news almost instantaneously before overshooting, particularly for large firms, consistent with faster information and algorithmic trading.

Table 6. Out-of-Sample Longevity of Post-Earnings-Announcement Drift

Cumulative abnormal returns (CAR) for high and low SUE portfolios

Holding Period (Trading Days)	Small Firms Diff (Hi-Lo)	Medium Firms Diff (Hi-Lo)	Large Firms Diff (Hi-Lo)
Postannouncement Drift (Cumulative Abnormal Returns, %)			
1 to 1	1.08	1.97	3.40
1 to 5	1.28	2.11	4.67
1 to 10	1.43	2.17	5.29
1 to 60	8.02	3.22	2.56
1 to 120	8.73	3.27	3.03
1 to 180	8.97	3.05	2.30
1 to 480	8.48	2.26	0.85
Postannouncement Drift (as a Fraction of 480-Day Drift)			
1 to 1	1.28	0.87	0.40
1 to 5	1.50	0.93	0.55
1 to 10	1.69	0.96	0.62
1 to 60	0.95	1.42	3.01
1 to 120	1.03	1.45	3.58
1 to 180	1.06	1.35	2.71
1 to 480	1.00	1.00	1.00

3.2.2 Test of risk premiums as explanation for the drift

a) Shifts in betas as a potential explanation

Pre-announcement *SUE-beta* rank correlation are near zero (0.15 and -0.10), versus 0.70 and 0.89 originally (Table 7). A significant positive correlation only appears in the (1,60) window and turns strongly negative by (121,180). Jensen's combined alpha remains economically meaningful at 4.6% (1,60), though not statistically significant. The original conclusion holds: *beta* shifts cannot explain PEAD's magnitude.

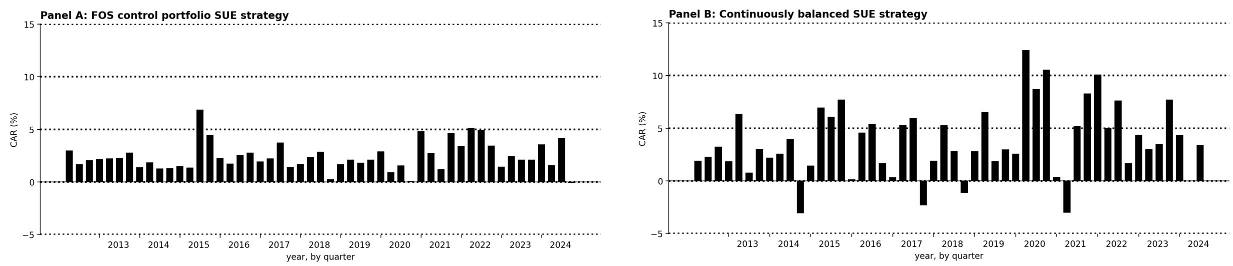
Table 7. OOS *Beta* Estimates by *SUE* Category, in Periods Surrounding Earnings Announcement

	Preannouncement Period		Postannouncement Period			
SUE Decile	(-119, -60)	(-59, 0)	(1, 60)	(61, 120)	(121, 180)	(181, 240)
Beta estimates						
1	1.39	1.44	1.34	1.51	1.54	1.33
2	1.51	1.32	1.23	1.39	1.41	1.30
9	1.37	1.28	1.38	1.33	1.30	1.14
10	1.52	1.32	1.34	1.42	1.34	1.42
Rank correlation, *SUE* & *beta*	0.15	-0.10	0.68*	-0.36	-0.87*	-0.15
Jensen's alpha						
$SUE = 1$	-5.0	-4.9	-3.4	-2.4	-1.6	-1.0
$SUE = 10$	0.6	1.3	1.1	-1.5	-1.8	-0.2
Combined	5.6	6.2	4.6	0.9	-0.2	0.8

b) Consistent profitability of the strategy

Strategy A generates positive returns in every out-of-sample quarter with only two near-zero quarters (Figure 5). B is more volatile (four negative quarters, four near-zero) but has a strong COVID period (8-12%). Both strategies remain profitable out-of-sample.

FIGURE 5. OOS cumulative abnormal returns (CARs) from *SUE* strategies, by calendar quarter.



3.2.3 Test of whether future earnings reflect full implication of past earnings

Quarter t *SUE* has little predictive power for t+1 earnings, with t-statistics below significance thresholds (Table 8). As a fraction of the 60-day drift, the predictable component is 7% for small firms and negligible for medium and large firms. This contrasts with the original 33-44%, suggesting that markets have become better at incorporating the serial correlation of quarterly earnings, even if short-term drift persists.

TABLE 8. OOS mean price reaction to quarter t+1 earnings (firms grouped on quarter t *SUE*)

	Abnormal Returns in [-4,0] Window of Q t+1 Earnings (in %) (t-values)		
SUE Decile for Quarter t	Small	Medium	Large
10 (good)	0.59 (3.21)	0.08 (0.60)	0.08 (0.90)
1 (bad)	0.09 (0.41)	0.19 (1.34)	0.09 (0.92)
Difference	0.60	-0.10	-0.01
As fraction of 60-day drift	7%	-3%	-0.4%

4 Conclusion

We closely replicated PEAD characteristics. Drift persists out-of-sample, with comparable positive returns. However, the structure of the drift has changed: markets now incorporate earnings news faster, with the entire long-run price adjustment for medium firms occurring within ten trading days. Predictability of t+1 returns disappeared, suggesting markets are more efficient at processing earnings' serial correlation.

5 References

Ball, S. P. Kothari, and R. Watts. "The Economics of the Relation Between Earnings Changes and Stock Returns." Working paper, University of Rochester, December 1988.

Ball, R and P. Brown. "An Empirical Evaluation of Accounting Income Numbers." Journal of Accounting Research (Autumn 1968): 159-78.

Chen, N., R. Roll, and S. Ross. "Economic Forces and the Stock Market." Journal of Business (July 1986): 383-404.

Foster, G. "Quarterly Accounting Data: Time Series Properties and Predictive-Ability Results." The Accounting Review (January 1977): 1-21.

Foster, G C. Olsen, and T. Shevlin. "Earnings Releases, Anomolies, and the Behavior of Security Returns." The Account ing Review (October 1984): 574-603.